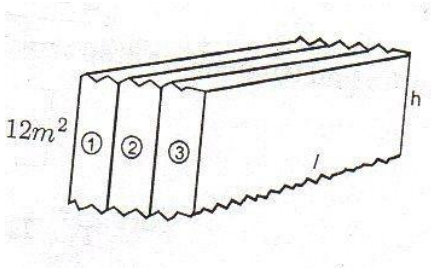
PTI 2206 Heat and mass Transfer: Calculations 2

1. **Calculate the rate of heat loss through the vertical walls of a boiler furnace of size 4 m by 3 m by 3 m high. The walls are constructed from an inner fire brick wall 25 cm thick of thermal conductivity 0.4 W/mK, a layer of ceramic blanket insulation of thermal conductivity 0.2 W/mK and 8 cm thick, and a steel protective layer of thermal conductivity 54 W/mK and 2 mm thick. The inside temperature of the fire brick layer was measured at 600° C and the temperature of the outside of the insulation 60^0 C. Also find the interface temperatures of layers.**

Solution

Given: Composite Wall l= 4m b= 3m h= 3m

Area of rectangular wall lb = 4x3 = 12m^2



L_1 = 25 cm Fire brick

K_1 = **0.4 W/mK**

L_2 = 0.08 m insulation

K_2 = **0.2 W/mK**

L_3 =0.002m Steel

K_3 = **54W/mK**

T_1 = 600$_0$ C, T_4 = 60^0 C

We know that,

$$Q = \frac{(\Delta T)_{overall}}{\Sigma R_{th}}$$

Here

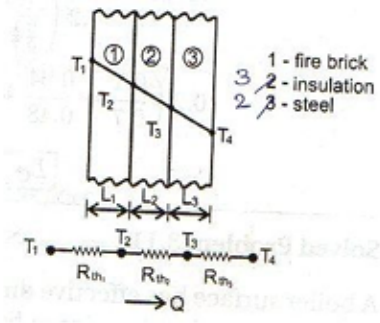
(ΔT) overall = $T_1 - T_4$

And $\Sigma R_{th} = R_{th1} + R_{th2} + R_{th3}$

$$R_{th1} = \frac{L_1}{k_1 A} = \frac{0.25}{0.4x12} = 0.0521 K/W$$

$$R_{th2} = \frac{L_2}{k_2 A} = \frac{0.08}{0.2x12} = 0.0333 K/W$$

$$R_{th3} = \frac{L_3}{k_3 A} = \frac{0.002}{54x12} = 0.0000031 K/W$$



$$Q = \frac{T_1 - T_4}{R_{th1} + R_{th2} + R_{th3}}$$

$$= \frac{600 - 60}{0.0521 + 0.0000031 + 0.0333}$$

$$Q = 6320.96 \text{ W}$$

From

$$Q = \frac{T_1 - T_4}{\sum R_{th}} = \frac{T_1 - T_2}{R_{th1}} = \frac{T_2 - T_3}{R_{th2}} = \frac{T_3 - T_4}{R_{th3}}$$

Thus

$$6320.96 = \frac{600 - T_2}{0.0521}, T_2 = 270.678 \text{ ℃ and}$$

$$6320.96 = \frac{T_3 - 60}{0.0000031}, T_3 = 60.0196 \text{ ℃}$$

2. A spherical container of negligible thickness holding a hot fluid at 140^0 and having an outer diameter of 0.4 m is insulated with three layers of each 50 mm thick insulation of k_1 = 0.02: k_2 = 0.06 and k_3 = 0.16 W/mK. (Starting from inside). The outside surface temperature is 30^0C. Determine (i) the heat loss, and (ii) Interface temperatures of insulating layers.

Given:

OD = 0.4 m

r_1 = 0.2 m

r_2 = r_1 + thickness of 1st insulation

= 0.2+0.05

r_2 = 0.25m

r_3 = r_2 + thickness of 2nd insulation

= 0.25+0.05

r_3 = 0.3m

r_4 = r_3 + thickness of 3rd insulation

= 0.3+0.05

r_4 = 0.35m

T_1 = 140° C, $T_2 = 30^o$ C,

k_1 = 0.02 W/mK

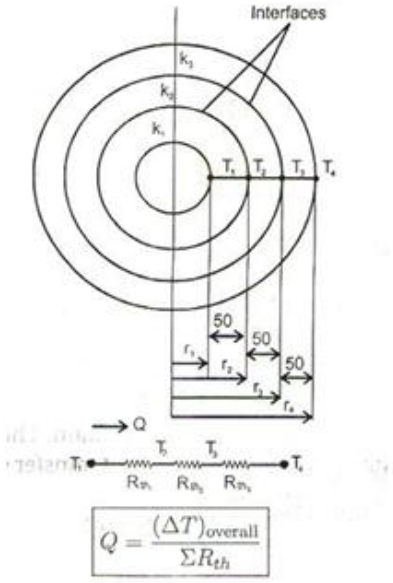
k_2 = 0.06 W/mK

k_3 = 0.16 W/mK.

Find (i) Q (ii) T_2, T_3

Solution

$$Q = \frac{(\Delta T)_{overall}}{\Sigma R_{th}}$$



$$Q = \frac{(\Delta T)_{overall}}{\Sigma R_{th}}$$

$\Delta T = T_{hf} - T_{cf}$

$\Sigma R_{th} = R_{th1} + R_{th2} + R_{th3}$

$$R_{th1} = \frac{r_2 - r_1}{4\pi k_1 r_2 r_1} = \frac{(0.25-0.20)}{4\pi x 0.02 x 0.25 x 0.2} = 3.978^\circ C/W$$

$$R_{th2} = \frac{r_3 - r_2}{4\pi k_2 r_3 r_2} = \frac{(0.30-0.25)}{4\pi x 0.06 x 0.3 x 0.25} = 0.8842^\circ C/W$$

$$R_{th1} = \frac{r_4 - r_3}{4\pi k_3 r_4 r_3} = \frac{(0.35-0.30)}{4\pi x 0.16 x 0.35 x 0.30} = 0.23684^\circ C/W$$

$$Q = \frac{140 - 30}{0.0796 + 0.8842 + 0.23684}$$

$Q = 21.57$ W

To find interface temperature (T_2, T_3)

$$Q = \frac{T_2 - T_3}{R_{th1}}$$

$T_2 = T_1 - [Q \times R_{th1}]$

$= 140 - [91.62 \times 0.0796]$

$T_2 = 54.17^0 C$

$$Q = \frac{T_2 - T_3}{R_{th1}}$$

$T_3 = T_2 - [Q \times R_{th2}]$

$= 132.71 - [91.62 \times 0.8842]$

$T_3 = 35.09^\circ C$

3. A steel tube with 5 cm ID, 7.6 cm OD and k=15W/m º C is covered with an insulative covering of thickness 2 cm and k 0.2 W/m ºC. A hot gas at 330º C with h = 400 W/m^{2o}C flows inside the tube. The outer surface of the insulation is exposed to cooler air at 30ºC with h = 60 W/m^{2o}C. Calculate the heat loss from the tube to the air for 10 m of the tube and the temperature drops resulting from the thermal resistances of the hot gas flow, the steel tube, the insulation layer and the outside air.

Given:

Inner diameter of steel, d_1 = 5 cm =0.05 m

Inner radius,r_1 = 0.025m

Outer diameter of steel, d_2 = 7.6 cm = 0.076m

Outer radius,r_2 = 0.038 m

Radius, $r_3 = r_2$ + thickness of insulation

= 0.038+0.02 m

r_3 = 0.058 m

Thermal conductivity of steel, k_1=15W/m º C

Thermal conductivity of insulation, k_2 = 0.2 W/m ºC

Hot gas temperature, T_{hf} = 330º C + 273 = 603 K

Heat transfer co-efficient at innear side, h_{hf} = 400 W/m^{2o}C

Ambient air temperature, T_{cf} = 30ºC +273 = 303 K

Heat transfer co-efficient at outer side h_{cf} = 60 W/m^{2o}C.

Length, L = 10 m

To find:

(i) Heat loss (Q)

(ii) Temperature drops ($T_{hf} - T_1$), ($T_1 - T_2$), ($T_2 - T_3$), ($T_3 - T_{cf}$),

Solution:

Heat flow $Q = \frac{\Delta T_{overall}}{\sum R_{th}}$

Where

$\Delta T_{overall} = T_{hf} - T_{cf}$

$$R = \frac{1}{2\pi L}\left[\frac{1}{h_{hf} r_1} + \frac{1}{k_1}\ln\left[\frac{r_2}{r_1}\right] + \frac{1}{k_2}\ln\left[\frac{r_3}{r_2}\right] + \frac{1}{k_3}\ln\left[\frac{r_4}{r_3}\right] + \frac{1}{h_{cf} r_4}\right]$$

$$Q = \frac{T_{hf} - T_{cf}}{\frac{1}{2\pi L}\left[\frac{1}{h_{hf} r_1} + \frac{1}{k_1}\ln\left[\frac{r_2}{r_1}\right] + \frac{1}{k_2}\ln\left[\frac{r_3}{r_2}\right] + \frac{1}{h_{cf} r_3}\right]}$$

$$Q = \frac{603 - 303}{\frac{1}{2\pi \times 10}\left[\frac{1}{400 \times 0.025} + \frac{1}{15}\ln\left[\frac{0.038}{0.025}\right] + \frac{1}{0.2}\ln\left[\frac{0.058}{0.038}\right] + \frac{1}{60 \times 0.058}\right]}$$

$Q = 7451.72$ W

We know that,

$$Q = \frac{T_{hf} - T_1}{R_{th\ conv.}}$$

$$= \frac{T_{hf} - T_1}{\frac{1}{2\pi L} \times \frac{1}{h_{hf} r_1}}$$

$$7451.72 = \frac{T_{hf} - T_1}{\frac{1}{2 \times \pi \times 10} \times \frac{1}{400 \times 0.025}}$$

$$T_{hf} - T_1 = 11.859K$$

$$Q = \frac{T_1 - T_2}{R_{th1}}$$

$$= \frac{T_1 - T_2}{\frac{1}{2\pi L} \times \left[\frac{1}{k_1}\ln\left[\frac{r_2}{r_1}\right]\right]}$$

$$7451.72 = \frac{T_1 - T_2}{\frac{1}{2 \times \pi \times 10} \times \frac{1}{15}\ln\left[\frac{0.038}{0.025}\right]}$$

$$T_1 - T_2 = 3.310\ K$$

$$Q = \frac{T_2 - T_3}{R_{th2}}$$

$$= \frac{T_2 - T_3}{\frac{1}{2\pi L} \times \left[\frac{1}{k_2}\ln\left[\frac{r_3}{r_2}\right]\right]}$$

$$7451.72 = \frac{T_2 - T_3}{\frac{1}{2 \times \pi \times 10} \times \frac{1}{0.2}\ln\left[\frac{0.058}{0.038}\right]}$$

$$T_2 - T_3 = 250.75\ K$$

$$Q = \frac{T_3 - T_{cf}}{R_{th\ conv.}}$$

$$= \frac{T_3 - T_{cf}}{\frac{1}{2\pi L} \times \frac{1}{h_{cf} r_3}}$$

$$7451.72 = \frac{T_3 - T_{cf}}{\frac{1}{2 \times \pi \times 10} \times \left[\frac{1}{60 \times 0.058}\right]}$$

$$T_3 - T_{cf} = 34.07K$$

4. **A spherical shaped vessel of 1.2 m diameter is 100 mm thick. Find the rate of heat leakage, if the temperature difference between the inner and outer surfaces is 200 ^oC. Thermal conductivity of material is 0.3 kJ /mh^oC.**

Given

d_1 =1.2 m r_1 = 0.6 m

r_2 = r_1 + thickness = 0.6 + 0.1

r_2 = 0.7 m

T_1-T_2=200^{0}C

K = 0.3 kJ /mhr ^oC = 0.0833 W/m^o C

Solution:

$$Q = \frac{\Delta T}{R_{th}} = \frac{T_1 - T_2}{R_{th}}$$

$$R_{th} = \frac{r_2 - r_1}{4\pi r_2 r_1} = \frac{(0.7-0.6)}{4\pi \times 0.0833 \times 0.6 \times 0.7} = 0.2275\ K/W$$

$$Q = \frac{\Delta T}{R_{th}} = \frac{200}{0.2275} = 879.132W$$

5. **A steel pipe (K = 45.0 W/m.K) having a 0.05m O.D is covered with a 0.042 m thick layer of magnesia (K = 0.07W/m.K) which in turn covered with a 0.024 m layer of fiberglass insulation (K = 0.048 W/m.K). The pipe wall outside temperature is 370 K and the outer surface temperature of the fiberglass is 305K. What is the interfacial temperature between the magnesia and fiberglass? Also calculate the steady state heat transfer.**

Given:

OD = 0.05 m d_1= 0.05 m r_1 = 0.025 m k_1 = 45 W/mK

r_2 = r_1 + thick of insulation 1 r_2 = 0.025+0.042 r_2 = 0.067 m k_2 = 0.07 W/mK

k_3 = 0.048 W/mK

r_3 = r_2 + thick of insulation 2

= 0.067+0.024 r_3 = 0.091 m

T_1 = 370 K , T_3 = 305 K

Here thickness of pipe is not given; neglect the thermal resistance of pipe.

$$Q = \frac{(\Delta T)overall}{\Sigma Rth}$$

Here

$$(\Delta T)overall = T_1 - T_3 = 370 - 305 = 65\ K$$

$$\Sigma R_{th} = R_{th1} + R_{th2}$$

$$R_{th1} = \frac{\ln\left(\frac{r_2}{r_1}\right)}{2\pi k_2 L} = \frac{\ln\left(\frac{0.067}{0.025}\right)}{2\pi \times 0.07 \times 1} = 2.2414 \text{ K/W}$$

$$R_{th2} = \frac{\ln\left(\frac{r_3}{r_2}\right)}{2\pi k_3 L} = \frac{\ln\left(\frac{0.091}{0.067}\right)}{2\pi \times 0.48 \times 1} = 1.0152 \text{ K/W}$$

$$Q = \frac{65}{2.2414 + 1.0152} = 19.959 \text{ W/m}$$

To find T_2

$$Q = \frac{T_1 - T_2}{R_{th1}}$$

$T_2 = T_1 - [Q \times R_{th1}]$

$= 370 - [19.959 \times 2.2414]$

$T_3 = 325.26K$

6. A mild steel tank of wall thickness 10 mm contains water at 90º C. The thermal conductivity of mild steel is 50 W/mºC , and the heat transfer coefficient for inside and outside of the tank area are 2800 and 11 W/m^2 ºC, respectively. If the atmospheric temperature is 20ºC , calculate

(a) The rate of heat loss per m^2 of the tank surface area.

(b) The temperature of the outside surface tank.

Given

L	=	10×10^{-3}m
T_{hf}	=	90 °C
k	=	50 W/m °C
h_{hf}	=	2800 W/m^2 °C
h_{cf}	=	11 W/m^2 °C
T_{cf}	=	20 ° C

Solution

$$Q = \frac{(\Delta T)overall}{\Sigma Rth}$$

Here $(\Delta T)_{overall} = T_{hf} - T_{cf} = 90 - 20 = 70^{o}C$

$$\sum R_{th} = R_{th_{conv_{hf}}} + R_{th1} + R_{th_{conv_{cf}}}$$

$$R_{th_{conv_{hf}}} = \frac{1}{h_{hf}.A} = \frac{1}{2800 \times 1} 0.00036\ K/W$$

$$R_{th} = \frac{L}{kA} = \frac{10 \times 10^{-3}}{50 \times 1} = 0.0002\ K/W$$

$$R_{th_{conv_{cf}}} = \frac{1}{h_{cf}.A} = \frac{1}{11 \times 1} 0.09091\ K/W$$

$$Q = \frac{70}{0.091469} = 765.29\ W/m^2$$

To find T_2

$$Q = \frac{T_{hf} - T_2}{R_{conv_{hf}} + R_{th\,1}}$$

$$T_2 = T_{hf} - \left[Q \times R_{conv_{hf}} + R_{th\,1}\right]$$

$$= 90 - [765 \text{x } 0.00056]$$

$$T_2 = 89.57\ ^{0}C$$

7. A 15 cm outer diameter steam pipe is covered with 5 cm high temperature insulation (k = 0.85 W/m ºC) and 4 cm of low temperature (k = 0.72 W/mº C). The steam is at 500 ºC and ambient air is at 40 ºC. Neglecting thermal resistance of steam and air sides and metal wall calculate the heat loss from 1000 mm length of the pipe. Also find temperature drop across the insulation.

Given

d_1	=	15 cm
r_1	=	$7.5 \text{ x} 10^{-2}$ m
r_2	=	r_1 + thick of high temperature insulation
r_2	=	$7.5 + 5 = 12.5 \text{ x } 10^{-2}$ m
r_3	=	r_2 + thick of low temperature insulation
r_3	=	$12.5 + 4 = 16.5 \text{ x } 10^{-2}$ m
k_{ins1}	=	0.85 w/m°C
$k_{ins\,2}$	=	0.72 w/m°C
T_{hf}	=	500 °C
T_{cf}	=	40 °C

To find

(i) Q if L = 1000mm = 1 m

Solution:

$$Q = \frac{(\Delta T)overall}{\Sigma Rth}$$

Here

$$\Delta T = T_1 - T_3$$

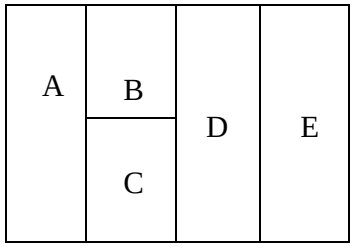
$$\Sigma R_{th} = R_{th1} + R_{th2}$$

$$R_{th1} = \frac{\ln\left(\frac{r_2}{r_1}\right)}{2\pi k_{1L}} = \frac{\ln\left(\frac{0.125}{0.075}\right)}{2\pi \times 0.85 \times 1} = 0.09564 \text{ K/W or } ^\circ \text{C/W}$$

$$R_{th2} = \frac{\ln\left(\frac{r_3}{r_2}\right)}{2\pi k_{2L}} = \frac{\ln\left(\frac{0.165}{0.125}\right)}{2\pi \times 0.72 \times 1} = 0.06137 \text{ K/W or } ^\circ \text{C/W}$$

$$Q = \frac{500 - 40}{0.09564 + 0.06137} = 2929.75 \text{W/m}$$

8. Determine the heat transfer through the composite wall shown in the figure below. Take the conductives of A, B, C, D & E as 50, 10, 6.67, 20 & 30 W/mK respectively and assume one dimensional heat transfer. Take of area of A =D= E = 1m^2 and B=C=0.5 m^2. Temperature entering at wall A is 800 ^{o} C and leaving at wall E is 100 ^{o} C.



Given:

$T_i = 800^\circ C$

$T_o = 100^\circ C$

$k_A = 50$ W/mK

$k_B = 10$ W/mK

$k_c = 6.67$ W/mK

$k_D = 20$ W/mK

$k_E = 30$ W/mK

$A_A = A_D = A_E = 1 m^2$

$A_B = A_C = 0.5 m^2$

Find

(i) Q

Solution

$$Q = \frac{(\Delta T)overall}{\Sigma Rth}$$

$$R_{th1} = R_{thA} = \frac{L_A}{k_A\, A}$$

Parallel $\frac{1}{R_{th2}} = \frac{1}{R_{thB}} + \frac{1}{R_{thC}} = \frac{R_{thB}+R_{thC}}{R_{thB}R_{thC}}$

$$R_{th2} = \frac{R_{thB}R_{thC}}{R_{thB} + R_{thC}}$$

$$R_{thB} = \frac{L_B}{k_B A_B}$$

$$R_{thC} = \frac{L_C}{k_C A_C}$$

$$R_{th4} = R_{thE} = \frac{L_E}{k_E A_E}$$

$$R_{th3} = R_{thD} = \frac{L_D}{k_D A_D}$$

$$R_{th1} = R_{thA} = \frac{1}{50 \times 1} = 0.02\ K/W$$

$$R_{thB} = \frac{1}{10 \times 0.5} = 0.2\ K/W$$

$$R_{thC} = \frac{1}{6.67 \times 0.5} = 0.2969\ K/W$$

$$R_{th2} = \frac{R_{thB}R_{thC}}{R_{thB} + R_{thC}} = \frac{0.2 \times 0.299}{0.2 + 0.299} = \frac{0.0598}{0.499}$$

$$R_{th2} = 0.1198\ K/W$$

$$R_{th3} = R_{thD} = \frac{L_D}{K_D A_D} = \frac{1}{20 \times 1} = 0.05\ K/W$$

$$R_{th4} = R_{thE} = \frac{L_E}{K_E A_E} = \frac{1}{30 \times 1} = 0.0333\ K/W$$

$$Q = \frac{T_i - T_o}{\sum R_{th}} = \frac{800 - 100}{0.02 + 0.1198 + 0.05 + 0.0333} = 3137.61W$$

$$Q = 3137.61W$$

Example 9

A 17-mm-diameter and 5-m-long electric wire is tightly wrapped with a 2-mm-thick plastic cover whose thermal conductivity is $k = 0.15$ W/m · °C. Electrical measurements indicate that a current of 10 A passes through the wire and there is a voltage drop of 8 V along the wire. If the insulated wire is exposed to a medium at $T_\infty = 30$°C with a heat transfer coefficient of $h = 12$ W/m² · °C, determine the temperature at the interface of the wire and the plastic cover in steady operation. Also determine whether doubling the thickness of the plastic cover will increase or decrease this interface temperature.

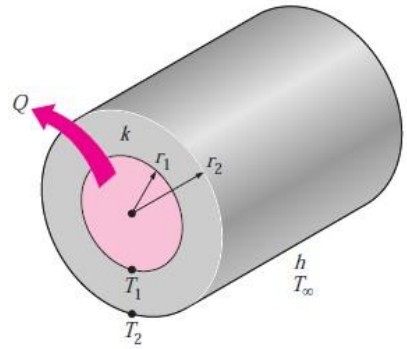
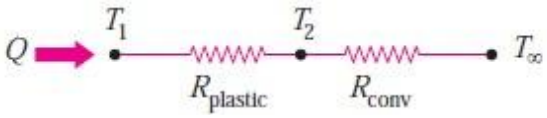


Figure Schematic for Example 9

Solution:



$$Q = \dot{W}_e = VI = (8\ \text{V})(10\ \text{A}) = 80\ \text{W}$$

$$A_2 = (2\pi r_2)L = 2\pi(0.0035\ \text{m})(5\ \text{m}) = 0.110\ \text{m}^2$$

$$R_{conv} = \frac{1}{hA_2} = \frac{1}{(12\ \text{W/m}^2 \cdot °\text{C})(0.110\ \text{m}^2)} = 0.76°\text{C/W}$$

$$R_{plastic} = \frac{\ln(r_2/r_1)}{2\pi kL} = \frac{\ln(3.5/1.5)}{2\pi(0.15\ \text{W/m} \cdot °\text{C})(5\ \text{m})} = 0.18°\text{C/W}$$

and therefore

$$R_{total} = R_{plastic} + R_{conv} = 0.76 + 0.18 = 0.94°\text{C/W}$$

Then the interface temperature can be determined from

$$Q = \frac{T_1 - T_\infty}{R_{total}} \longrightarrow T_1 = T_\infty + QR_{total} = 30°\text{C} + (80\ \text{W})(0.94°\text{C/W}) = 105°\text{C}$$

$$r_{cr} = \frac{k}{h} = \frac{0.15\ \text{W/m} \cdot °\text{C}}{12\ \text{W/m}^2 \cdot °\text{C}} = 0.0125\ \text{m} = 12.5\ \text{mm}$$

Example 10

(a) Explain the term critical radius of insulation.

The **critical radius of insulation** refers to the outer radius of insulation at which the heat loss from an insulated cylindrical or spherical object is maximized. Beyond this radius, adding more insulation actually reduces heat loss, while below it, adding insulation increases heat loss.

(b) An electrical wire has a diameter of 2 mm and is exposed to air with a convective heat transfer coefficient of 20 W/m²·K. To prevent overheating, it is coated with an insulating material of thermal conductivity 0.1 W/m·K.

(i) Calculate the critical radius of insulation.

$$r_{cr} = \frac{k}{h} = \frac{0.1}{20} = 0.005\,\text{m} = 5\,\text{mm}$$

(ii) Determine if applying a 1 mm thick insulation layer will increase or decrease heat transfer.

- **Bare wire radius (r)** = 1 mm
- **Insulation thickness** = 1 mm → **Total radius after insulation (r_{ins})** = 1 + 1 = **2 mm**
- **Critical radius (r_{cr})** = 5 mm

Since $r_{ins} = 2\,\text{mm} < r_{cr} = 5\,\text{mm}$, adding insulation **increases heat transfer**.

Applying a 1 mm insulation layer increases heat transfer because the total radius (2 mm) is less than the critical radius (5 mm).

(iii) Find the minimum insulation thickness required to start reducing heat loss.

To **reduce heat loss**, the total radius must exceed the critical radius:

$$r_{ins} > r_{cr}$$

$$\text{Insulation thickness} > r_{cr} - r = 5\,\text{mm} - 1\,\text{mm} = 4\,\text{mm}$$

The minimum insulation thickness needed to reduce heat loss is 4 mm.

Exercise

(a) A steel tube with 5 cm ID, 7.6 cm OD and k=15W/m^{o}C is covered with an insulative covering of thickness 2 cm and thermal conductivity of 0.2 W/m^{o}C. A hot gas at 330 ^{o}C with convective heat transfer coefficient of 400 W/m^{2o}C flows inside the tube. The outer surface of the insulation is exposed to cooler air at 30 ^{o}C with convective heat transfer coefficient of 60 W/m^{2o}C. Calculate the heat loss from the tube to the air for 10 m of the tube and the temperature drops resulting from the thermal resistances of the hot gas flow, the steel tube, the insulation layer and the outside air.

(b) A spherical tank is of an internal and external diameter 4 m and 4.04 m respectively. The tank material is stainless steel of thermal conductivity 16 W/m^{2}K. The tank contains iced water at 0 ^{o}C and located in a room of temperature 23 ^{o}C. The heat transfer coefficients in and out the sphere is 90 W/m^{2}K and 8 W/m^{2}K respectively. Determine the heat transfer rate to the iced water in the tank.

(c) A spherical container of negligible thickness holding a hot fluid at 140 ^{o}C and having an outer diameter of 0.4 m is insulated with three layers of each 50 mm thick insulation of k_1 = 0.02: k_2 = 0.06 and k_3 = 0.16 W/mK. (Starting from inside). The outside surface temperature is 30 ^{o}C. Determine;

(i) the heat loss,

(ii) Interface temperatures of insulating layers.